

Abdullah Nayem Wasi Emran

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RESEARCH INTEREST

Computational Biology & Bioinformatics, AI/ML for Healthcare, Biological Foundation Models, Causal & Fair Machine Learning

EDUCATION

MSc in Computer Science and Engineering (CSE)

May 2025 - Mar 2027 (Expected)

Bangladesh University of Engineering and Technology (BUET)

Thesis: *Biological Novelty and Generalization in RNA-Protein Interaction Prediction*

BSc in Computer Science and Engineering (CSE)

Feb 2020 - Mar 2025

Bangladesh University of Engineering and Technology (BUET)

Dean's List Award and University Merit Scholarship in four of eight semesters.

PUBLICATIONS

Journal Articles

A. N. W. Emran and A. B. M. A. Al Islam, "Domain adaptive uplift modeling across heterogeneous mental health cohorts," *iScience*, vol. 29, no. 6, art. 116027, 2026. [DOI](#)

A. N. W. Emran, M. J. Disha, and R. Reaz, "Fusing exploration and exploitation: hybrid fireworks-whale optimization for multi-objective independent task scheduling in cloud environments," *Cluster Computing*, vol. 29, no. 9, art. 533, 2026. [DOI](#)

A. N. W. Emran and A. B. M. A. Al Islam, "Mental health across contexts: a cross-dataset study covering medical students, quarantined individuals, and psychiatric disordered subjects," *Humanities and Social Sciences Communications*, vol. 12, art. 1157, 2025. [DOI](#)

Conference & Workshop Papers

A. N. W. Emran and M. S. Rahman, "When does graph refinement help? A controlled study of unseen-molecule RNA-protein interaction prediction," *PM4B Workshop, IEEE ICDM*, 2026 (accepted).

A. N. W. Emran and T. Rahman, "SCOPEs: Stability-aware cross-platform feature selection for matched TCGA gene expression and RNA-Seq data," *AAAI 2026 Workshop on AI to Accelerate Science and Engineering (AI2ASE)*, 2026.

RESEARCH EXPERIENCE

Biological Novelty and Generalization in RNA-Protein Interaction Prediction (MSc Thesis)

Supervised by Dr. Mohammad Saifur Rahman

2025 - present

(*Bioinformatics, Biological Foundation Models, Graph Neural Networks*)

- Built a controlled framework that moves from exact cold start to sequence-family holdout and independent external evaluation; verified leakage-free exact splits (60/60 checks) and found added NPInter2 protein-family difficulty (AUROC 0.906 $\rightarrow$ 0.877).
- Benchmarked 21 RNA/protein foundation-model combinations, with reproducible rankings across three family-held partitions and strong cross-modal representation effects.
- Found 59.3% of TheNovel interactions family-novel on both molecular sides (PairMLP-Cross AUROC 0.734 vs. 0.832 overall); graph topology provided context-dependent rather than universal gains. Earlier stage accepted at PM4B, IEEE ICDM 2026.

Failure-Risk Estimation for RNA-Protein Interaction Prediction

Supervised by Dr. Mohammad Saifur Rahman and Dr. A. B. M. Alim Al Islam

2026 - present

(*Reliability, Out-of-Distribution Detection, Selective Prediction*)

- Developed a leakage-aware framework that estimates failure risk for individual protein-cold RPI predictions using decision certainty, RNA training support, and embedding familiarity.
- Across four RPI systems and three benchmarks (180 runs), protein-disjoint evaluation reduced apparent failure AUROC by 0.094–0.262; engineered features improved failure AUROC by up to 0.240. Manuscript in preparation.

Domain-Adaptive Uplift Modeling across Heterogeneous Mental-Health Cohorts

Supervised by Dr. A. B. M. Alim Al Islam

Apr 2025 - Jan 2026

(Causal Inference, Domain Adaptation, Fair Machine Learning)

- Harmonized student, quarantine, and clinical cohorts and trained a domain-adversarial (DANN) uplift model with fairness-aware regularization.
- Evaluated generalization with leave-one-cohort-out AUUC and studied cohort-specific calibration for unstable treatment effects. Published in *iScience*.

Mental Health Across Contexts: A Cross-Dataset Study

Supervised by Dr. A. B. M. Alim Al Islam

Jul 2024 - Jan 2025

(Data Mining, Network Analysis, Causal Inference)

- Analyzed mental-health outcomes across medical-student, quarantine, and psychiatric cohorts using statistical modeling, propensity score matching, and network analysis.
- Identified robust and context-dependent cross-dataset patterns. Published in *Humanities and Social Sciences Communications*.

Hybrid Fireworks-Whale Optimization for Multi-Objective Cloud Task Scheduling (BSc Thesis)

Supervised by Dr. Rezwana Reaz

Nov 2023 - Mar 2025

(Cloud Computing, Optimization)

- Designed sequential, parallel, and component-level FWA-WOA hybrids for multi-objective cloud task scheduling.
- Benchmarked makespan, cost, and resource utilization on synthetic workloads and GoCJ traces, achieving 2.5-7.5% higher composite fitness than baselines. Published in *Cluster Computing*.

WORK EXPERIENCE

Lecturer (Full-time) - Department of CSE, BRAC University

June 2025 - present

- Cybersecurity (CSE490B, theory/lab); Bioinformatics (CSE443)
- Computer Networks (CSE421, theory/lab); Discrete Mathematics (CSE230)
- Algorithms Lab (CSE221); Compiler Design (CSE420)

SELECTED PROJECTS

SCOPES: Cross-Platform Gene Signature Selection – leak-free NSGA-II pipeline optimizing AUC, Kuncheva stability, and MMD for cross-platform TCGA-BRCA gene selection. [Code](#)

Lightweight Histopathological Cancer Cell Classification – EfficientNet-B0 with SE/MSFF/CBAM and PathMNIST transfer learning; 98% accuracy on CRC-VAL-HE-7K. [Code](#)

HyMAGNN: Lightweight Hypergraph GNN for Multivariate Time-Series Forecasting – hypergraph modeling, deformable convolution, and feature fusion for short-range multivariate forecasting. [Code](#)

SKILLS

Programming: Python (PyTorch, Scikit-learn, Pandas), C/C++, Java, JavaScript, Bash, SQL.

Tools: Git, L^AT_EX, NS-3, OpenGL, Wireshark, Autopsy, Next.js, Django, Oracle DBMS.

AWARDS & HONORS

Dean's List Award and University Merit Scholarship - BUET, four of eight undergraduate semesters 2022, 2023

REFERENCES

Dr. Mohammad Saifur Rahman Professor, CSE, BUET MSc Thesis Supervisor mrahman@cse.buet.ac.bd	Dr. A. B. M. Alim Al Islam Professor, CSE, BUET Research Supervisor alim_razi@cse.buet.ac.bd	Dr. Rezwana Reaz Associate Professor, CSE, BUET BSc Thesis Supervisor rampi@cse.buet.ac.bd
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